

STAREHE GIRLS' 2025 PREDICTION



POSSIBLE KCSE EXAM

231/3

BIOLOGY

PAPER 3 (PRACTICAL)

TIME: 1¾ HOURS



NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO CANDIDATES:

- Write your **name** and **admission number** in the spaces provided.
- Sign and write **date** of examination in the spaces provided above
- Answer **all** the questions in section **in the spaces** provided after each question only
- You are required to spend the first 15 minutes of the 1 ¾ hours allowed for this paper, reading the whole paper carefully before starting to do this paper.

For Examiner's Use Only:

QUESTIONS	MAXIMUM SCORE	CANDIDATE'S SCORE
1	14	
2	14	
3	12	
TOTAL	40	

1. You are provided with olive oil, liquids labeled **L₁** and **L₂**, and an Irish potato.

Label two test tubes **A** and **B**.

Place 2 cm³ of water in each test tube.

Add 8 drops of olive oil into each test tube.

To test tube labelled **A** add 8 drops of liquid **L₁**

Shake both test tubes. Allow to stand for 2 minutes.

(a)i) Record your observations.

(2 Marks)

Test tube **A**.

.....
.....

Test tube **B**.

.....
.....

(ii) Name the process that has taken place in test tube **A**.

(1 Mark)

.....

(iii) State the significance of the process named in **(a)(ii)** above.

(1 Mark)

.....
.....

(iv) Name the:

Digestive juice in humans that has the same effect on oil as liquid **L₁**.

(1 Mark)

.....

Region of alimentary canal into which the juice is secreted.

(1 Mark)

.....

(b) Label two test tubes **C** and **D**.

Place 2 cm³ of liquid L₂ into each test tube. Add **one** drop of iodine solution into each test tube.

(i) Record your observation.

(1 Mark)

.....
.....

(ii) What is the identity of L₂?

(1 Mark)

.....

Cut a cube whose sides are 1cm from the Irish potato provided. Crush the cube to obtain a paste. Place the paste in the test tube labeled C. Leave the set up for at least 30 minutes.

(iii) Record your observation in C and D

(2 Marks)

C
D

(iv) **Account** for the results in (b)(iii) above.

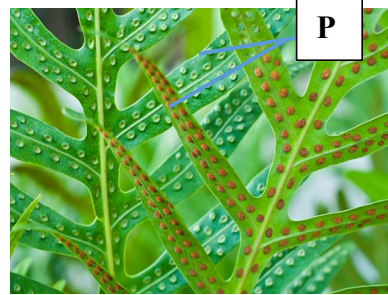
(4 Marks)

C.....
.....
...
D.....
.....
...

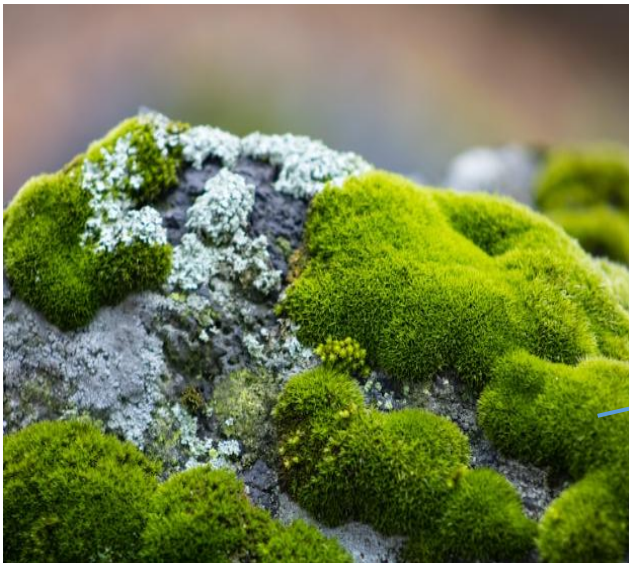
2. Below are photographs of specimens obtained from plants. Examine the photographs and use them to answer the questions that follow.



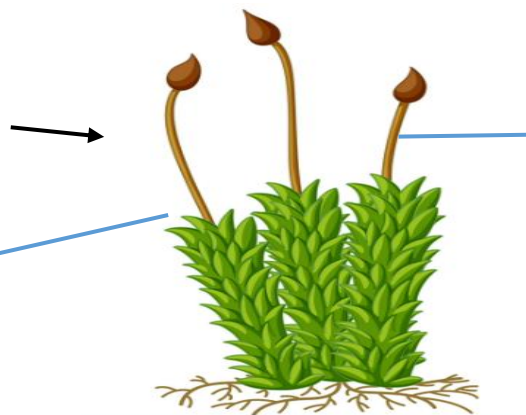
A



P



B



Q

(a). Name the plants and state the divisions in which these specimens belong
(Marks)

(4

A

Name.....

Division.....

B

Name.....

Division.....

(b). On the diagrams, name structure labelled **P** and **Q**
(2 Marks)

(c). State **two** similarities and two differences of the organisms represented by the images above.

Similarities

(2 Marks)

.....

.....

.....

.....

Differences

(2 Marks)

.....

.....

.....

.....

(d). The organisms shown above shows alternation of generation.

(i) Explain what is meant by the concept alternation of generation.

(1 Mark)

.....

.....

(ii) Which is the dominant generation in the organisms represented by image **A**

(1 Mark)

.....

.....

(e). Name the **two** structures that produce gametes when the two-organisms shown above are reproducing by means of gametes.

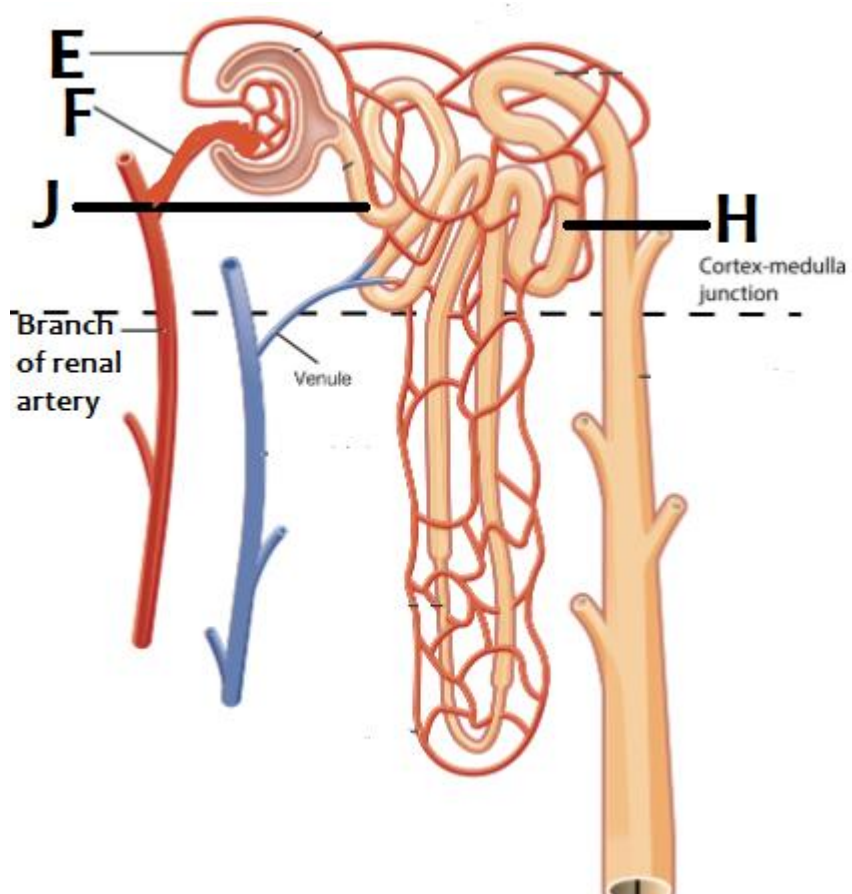
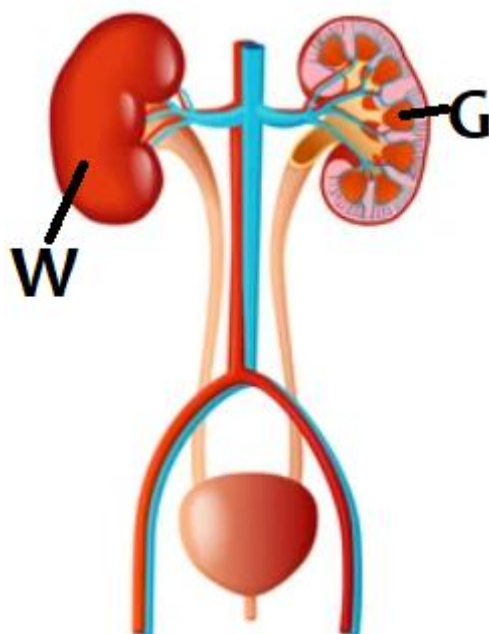
(2 Marks)

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.....

.....

3. The images below were obtained from certain system in a mammal. Study the images and use them to answer the questions that follow.



(a) Name the part labeled **W** and state its functions.

Name

(1 Marks)

.....

Functions

(2 Marks)

.....

.....

.....

(b). State **one** difference between vessel **E** and **F** and explain its importance.

Difference

(1 Mark)

.....

.....

.....

Importance

(1 Mark)

.....

.....

(c). (i) Name **two** substances reabsorbed in part labelled **J**

(2 Marks)

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.....

.....

(ii) Identify the physiological process involved in reabsorption of substances in part **H** (1 Mark)

.....

.....

(iii) Explain **four** adaptations of part **H** and **J** to its functions

(4 Marks)

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